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**1sem BCA sec A**

**Day 3: Assessment task1**

**Q1) What is data science, really?**

Data science is a totally a different thing which works in converting a real-world huge data into a decision that easily works on.

It works with its tools i.e., AI, ML, DL depending on the requirements.

Eg: finding patterns, making decisions

**Q2) What are the steps in the data science loop?**

The steps of data science are:

**Question:** defining the thing which we are trying to solve.

**Data:** collecting or gathering the required data to that question.

**Model**: structuring or building a model or analyzing pattern and answering the question.

**Check-back:** evaluating the results whether the answer is valid or not.

**Decision:** using all these steps to decide or take action.

**Q3) Does data science always need AI?**

No, data science does not need AI always as it works on the problem and gives the correct decisions without AI. It works more effectively in cleaning data, exploring it, building dashboards, A/B testing.

**Q4) Find the decision and the check-back**

**Decision:** The canteen decides to prepare fewer sandwiches next Friday, based on the data showing many go unsold every Friday.

**Check-back**: The canteen counts the leftover sandwiches again after that Friday and compares them with the earlier numbers to see whether the waste went down. If it did, the decision worked. If sandwiches ran out too early or plenty were still left, they adjust the number and repeat the loop.

**Q5) find data science without AI**

**Is this data science?**

Yes

**What decision did the shopkeeper make?**

The shopkeeper decided to stock more umbrellas before the next rainy day, based on the pattern that more umbrellas sell when it rains.

**How did the shopkeeper check whether the decision worked?**

The shopkeeper compared the new umbrella sales with the old sales recorded in the notebook. If sales went up and few umbrellas were left over or missed, the decision worked. If not, the stock amount can be adjusted and the loop repeated.

**Q6) Complete the missing part of the loop**

**Missing step:** Check-back. The college made a decision (moving the class) but never measured the result afterward.

**Action**: The college should record attendance after the change and compare it with the earlier attendance numbers to see whether it actually improved.

**Why it matters:** Without check-back, the college can't know if the decision worked. Attendance may have stayed low for other reasons, such as the subject, the teacher, or student schedules. Checking back tells them whether to keep the change or try something else, which is what completes the loop.

**Q7) Build a simple data science loop**

**Pattern**: The student leaves home at 8:15 a.m. and reaches college late on many days, which suggests the departure time is too late for the travel time involved.

**Decision:** The student will try leaving 20 minutes earlier, at 7:55 a.m., for the next two weeks.

**Check-back**: The student should record the arrival time at college each day (and whether it was on time or late), then compare these with the earlier late-arrival records. If late arrivals dropped, the change worked. If not, the student can adjust the departure time again or look at other causes like traffic or transport delays.